NUO (IVY) LIU, PH.D.

Postdoctoral Fellow | Goodarzi Lab, Arc Institute & University of California, San Francisco
ivy.liu@arcinstitute.org | nuoliu.github.io | Brooklyn, NY

Computational biologist working across single-cell genomics, protein design, perturbation screening, and LLMs for biological reasoning; PhD from MIT with experience co-designing experimental frameworks with wet-lab collaborators.

Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA

Ph.D., Computational and Systems Biology | GPA: 4.9/5.0 | Sep 2020 – May 2025
Advisor: Prof. Alex K. Shalek (Chemistry, IMES, HST)
Thesis Advisory Committee: Prof. Forest White, Prof. Ernest Fraenkel

Harvey Mudd College — Claremont, CA

B.S., Mathematical and Computational Biology | GPA: 3.95/4.0 | Aug 2016 – May 2020
Graduated with High Distinction | Honors in Biology | Honors in Computer Science
Humanities concentration in Economics | Dean's List all applicable semesters

Honors & Awards

David H. Koch Graduate Fellowship, Koch Institute, MIT	2023–24
Computer Science Class of '94 Award, Harvey Mudd College	2020
CRA Outstanding Undergraduate Research Award, Finalist	2020
W.A. Brandenburger Biology Prize, Harvey Mudd College	2019
First-Tier Traveling Fellowship, 17th Asia Pacific Bioinformatics Conference	2019
CRC Handbook Prize, Harvey Mudd College	2017

Technical Skills

Languages: Python, R, Bash, SQL
ML/AI: PyTorch, distributed training (DDP/SLURM), Hugging Face, foundation model training (RNA/sequence models), benchmark and eval design for biological reasoning models
Single-cell & genomics: Cell Ranger, FastQC, CellBender, scverse (scanpy, anndata, muon), Seurat, scVI, Harmony, pySCENIC, LIANA, pertpy, VISION, DESeq2/pydeseq2, bedtools; active collaborator with Arc's internal single-cell analysis pipeline engineering team
Structural & protein design: RFdiffusion, ProteinMPNN, Boltz2, AlphaFold
Wet lab: Sample processing for single-cell RNA-seq (10x Genomics protocols, Seq-Well S3), cell-culture (cancer cell lines, intestinal organoids)
Infrastructure: SLURM/HPC, GCP, tmux, Claude Code, MCP tool development

Research Experience

Arc Institute — Postdoctoral Fellow, Goodarzi Lab — Aug 2025 – Present
Advisor: Prof. Hani Goodarzi (Arc/UCSF)
Projects spans model development, protein design, and functional genomics — ranging from co-scoping experimental design with wet-lab collaborators to leading downstream analyses of collaborator-generated data.

- Developing cancer-specialized RNA foundation models for oncRNA target discovery across 32 cancer types; Building hybrid computational-experimental pipelines for RNA 3D structure prediction; Engineering AI-driven synthetic RNA-binding protein (RBP) switches for precision targeting of cancer-specific oncRNAs
- Developing multimodal biological reasoning LLMs for protein function prediction (BioReason-Pro)
- Developing the STATE virtual cell model on a Gene × Environment (G×E) map spanning 40+ cancer cell lines and 96 metabolic perturbations to predict cancer metabolic vulnerabilities in silico; co-scoping validation experiments with Jain Lab and Tahoe Therapeutics (Perturb-Env)

- Mentoring a project on multimodal latent causal VAE for joint inference of gene regulatory and protein interaction networks (Goudarzi, Liu, Goodarzi, MLGenX workshop ICML 2026).
- Led scRNA-seq characterization of microglia APOE variant effects on Alzheimer's disease in human *ex vivo* models with Konermann Lab; analysis-driven hypotheses now driving follow-up Perturb-seq screens (in collaboration with Konermann Lab
- Characterizing ENPP1 and STING mutation effects on mouse brain inflammations using scRNA-seq data (in collaboration with Li Lab)

Massachusetts Institute of Technology — Graduate Researcher, Shalek Lab Jun 2021 – Aug 2025

Broad Institute & Ragon Institute, Cambridge, MA

- Co-led design of compressed phenotypic screening from experimental protocol through computational analysis: designed the PBMC screen end-to-end with co-first author and coordinated screening design with the Broad Institute CDoT team; discovered pleiotropic effects of JAK–STAT inhibitors on human immune cells. Framework broadly applicable to *ex vivo* multicellular models. Code: github.com/ShalekLab/compressed_screening
- Led single-cell and spatial transcriptomic profiling of human TB-diseased lung tissue; identified myofibroblast-like populations as central drivers of TB pathogenesis and lung remodeling; work redefined TB fibrosis biology
- Used single-cell transcriptomics to characterize functional mechanisms of long-term growth hormone administration effects on MASLD/NAFLD patients

Westlake University — Summer Research Intern, Guo Lab Jul – Aug 2020

Hangzhou, China | Advisor: Dr. Tiannan Guo

- Built and trained an LSTM-based deep learning model on pan-human tissue proteomics to predict ion mobility values (Pearson's $r > 0.98$)
- Predicted ion mobility values for mouse liver proteomics data from non-timsTOF instruments; compared library search results using ion mobility-supplemented libraries

Harvey Mudd College — Senior Thesis Researcher, Bush Lab Sep 2019 – May 2020

Claremont, CA | Advisor: Prof. Eliot Bush

- Year-long thesis: Reconstructing Gene Family Evolution in Microbes Using the DTLOR Algorithm
- Extended Duplication-Transfer-Loss model to include new syntenic events for more comprehensive reconstruction of prokaryotic evolutionary history

Baylor College of Medicine — Summer Research Intern, Sumazin Lab Summer 2019

Houston, TX | Advisor: Dr. Pavel Sumazin

- Designed a method to estimate cell-type specific expression from bulk tumor profiles using a hierarchical Linear Programming approach to reconstruct transcriptional evolution of tumors

Baylor College of Medicine — Summer Research Intern, Coarfa Lab Summer 2018

Houston, TX | Advisor: Dr. Cristian Coarfa

- Performed RNA-seq, microarray, and gene enrichment analysis on rat liver epigenome under environmental influences
- Built and trained a CNN to predict epigenetic remarking events; extracted de novo DNA motifs related to epigenetic reprogramming

Harvey Mudd College — Research Intern, Libeskind-Hadas & Wu Labs Summer 2017

Claremont, CA

- Extended dynamic programming algorithm for phylogenetic tree reconciliation under the Duplication-Loss-Coalescence model to optimize Pareto-optimal solutions

Harvey Mudd College — Research Intern, McFadden Lab Spring 2017

Claremont, CA | Advisor: Prof. Catherine McFadden

- Investigated feasibility of 28S rDNA genetic barcoding to differentiate species within coral genus Sinularia; demonstrated promise as an alternative to whole-genome sequencing

Publications & Preprints

** Co-first authorship*

2026

Ghirardelli Smith O.C.*, Dao T.T.*, Gavil N.V.*, O'Flanagan S.D., Rubin A.J., Nguyen S., Watowich M.B., **Liu, N.**, Weyu E., Quarnstrom C.F., Soerens A.G., Joag V., Rosato P.C., Krummel M.F., Geller M.A., Miller J.S., Giubellino A., Vezys V., Shalek A.K., Masopust D. (2026). Robust T cell activation kills tumors regardless of T cell specificity. *Nature Immunology* (accepted).

Fallahpour, A., Seyed-Ahmadi, A., Idehpour, P., Ibrahim, O., Gupta, P., Naimer, J., Zhu, K., Shah, A., Ma, S., Adduri, A., Güloğlu, T., **Liu, N.**, Cui, H., Jain, A., de Castro, M., Stiles, J.S., Nemcko, F., Nevue, A.A., Moon, H.C., Sosnick, L., Markham, O., Duan, H., Lee, M.Y.Y., Salvador, A.F.M., Maddison, C.J., Thaiss, C.A., Ricci-Tam, C., Plosky, B.S., Burke, D.P., Hsu, P.D., Goodarzi, H., Wang, B. (2026). BioReason-Pro: Advancing Protein Function Prediction with Multimodal Biological Reasoning. *bioRxiv* 10.64898/2026.03.19.712954.

Liu, N.*, Mbano, I.M.*, Wadsworth, M.H., Chambers, M.J., Mpotje, T., Asowata, O.E., Nyquist, S.K., Nargan, K., Ramsuran, D., Karim, F., Hughes, T., Bromley, J.D., Krause, R., Naidoo, T., Tezera, L., Reichmann, M.T., Ganchua, S.K., Kløverpris, H.N., Dullabh, K.J., ... Leslie, A. (2026). Single-cell and spatial profiling highlights TB-induced myofibroblasts as drivers of lung pathology. *J. Exp. Med.* 223(3), e20251067. https://doi.org/10.1084/jem.20251067 (Front cover)

Established myofibroblasts as central drivers of TB-induced lung pathology using single-cell and spatial transcriptomics of human TB lung tissue.

2025

Liu, N.*, Kattan, W.E.*, Mead, B.E.*, Kummerlowe, C.*, Cheng, T., Cheah, J.H., Soule, C.K., Peters, J., Lowder, K.E., Blainey, P.C., Hahn, W.C., Cleary, B., Bryson, B., Winter, P.S., Raghavan, S., & Shalek, A.K. (2025). Scalable, compressed phenotypic screening using pooled perturbations. *Nature Biotechnology* 43, 1324–1336. https://doi.org/10.1038/s41587-024-02403-z

*Introduced an experimental + computational framework for scaling phenotypic drug screens on ex vivo multicellular models. **Systems:** patient-derived PDAC organoids, primary human PBMCs. **Perturbations:** recombinant TME ligand libraries, annotated small-molecule libraries. **Readouts:** scRNA-seq, high-content Cell Painting imaging. Identified pleiotropic effects of JAK–STAT inhibitors on human immune cells. Featured in MIT News (news.mit.edu/2024/new-framework-efficiently-screen-drugs-1017); accompanying Nature Biotechnology Research Briefing (doi.org/10.1038/s41587-024-02404-y).*

2023

Liu, N., Gonzalez, T.A., Fischer, J., Hong, C., Johnson, M., Mawhorter, R., Mugnatto, F., Soh, R., Somji, S., Wirth, J.S., Libeskind-Hadas, R., & Bush, E.C. (2023). xenoGI 3: Using the DTLOR model to reconstruct the evolution of gene families in clades of microbes. *BMC Bioinformatics* 24, 295. https://doi.org/10.1186/s12859-023-05410-0. Code: github.com/ecbush/xenoGI

2021

Raghavan, S., Winter, P.S., Navia, A.W., Williams, H.L., DenAdel, A., Lowder, K.E., Galvez-Reyes, J., Kalekar, R.L., Mulugeta, N., Kapner, K.S., Raghavan, M.S., Borah, A.A., **Liu, N.**, Väyrynen, S.A., Costa, A.D., Ng, R.W.S., Wang, J., Hill, E.K., Ragon, D.Y., … Shalek, A.K. (2021). Microenvironment drives cell state, plasticity, and drug response in pancreatic cancer. *Cell* 184(25), 6119–6137. https://doi.org/10.1016/j.cell.2021.11.017

Du, H., Ong, Y.S., Knittel, M., Mawhorter, R., **Liu, N.**, Gross, G., Tojo, R., Libeskind-Hadas, R., & Wu, Y.-C. (2021). Multiple Optimal Reconciliations Under the Duplication-Loss-Coalescence Model. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 18, 2144–2156. https://doi.org/10.1109/TCBB.2019.2922337

Liu, J., Mawhorter, R., **Liu, N.**, Santichaivekin, S., Bush, E., & Libeskind-Hadas, R. (2021). Maximum parsimony reconciliation in the DTLOR model. *BMC Bioinformatics* 22, 394. https://doi.org/10.1186/s12859-021-04290-6

2020

Carothers, M., Gardi, J., Gross, G., Kuze, T., **Liu, N.**, Plunkett, F., Qian, J., & Wu, Y.-C. (2020). An Integer Linear Programming Solution for the Most Parsimonious Reconciliation Problem under the Duplication-Loss-Coalescence Model. *ACM BCB 2020.* https://doi.org/10.1145/3388440.3412474

2019

Mawhorter, R., **Liu, N.**, Libeskind-Hadas, R., & Wu, Y.-C. (2019). Inferring Pareto-optimal reconciliations across multiple event costs under the duplication-loss-coalescence model. *BMC Bioinformatics* 20, 639. https://doi.org/10.1186/s12859-019-3206-6

Selected Presentations

Lightning Talk (Nominated): "Perturb-Env: Towards a GxE map for virtual cells" — Feb 2026
Arc Institute Symposium, San Francisco, CA

Poster: "Single-cell profiling reveals a role for tuberculosis-induced myofibroblasts in the immunopathology of infected lungs" — May 2025
AAI Annual Meeting (Immunology 2025), Honolulu, HI

Selected Talk: "Scalable, compressed phenotypic screening using pooled perturbations" — Feb 2025
Winter q-bio Conference 2025, Ko Olina, HI

Oral: Compressed phenotypic screening empowers scalable biological discovery — May 2024
Single-Cell Genomics Conference GRC, Les Diablerets, Switzerland

Talk: Compressed Screening — High-throughput measurement and perturbation of tissues — Feb 2024
Microsoft Research New England, Cambridge, MA

Talk: Putative drivers of cellular plasticity in glioblastoma — Jul 2022
NCI Human Tumor Atlas Network (HTAN) Internal Meeting, Cambridge, MA

Teaching & Service

Mentor — AIxBio fellow program, Arc Institute — Spring 2026-Current

Teaching Assistant — Modern Biostatistics (7.093) & Modern Computational Biology (7.094), MIT — Spring 2022
Designed and led weekly recitations and office hours; co-developed course materials; graded problem sets and exams

Teacher — MIT Spark 2022 (Immunology for 7th & 8th graders) — Mar 2022

Workshop Organizer — Single-Cell Genomics Workshop in Asia, Human Cell Atlas 2024

Executive Board (Secretary & Fund Manager) — CSB Application Assistance Program, MIT 2021 – 2025

Helped establish the program; secured Grad Student Experience Grant from MIT Office of Graduate Education

Co-President & Director of Planning — MIT-CHIEF (China Innovation & Entrepreneurship Forum) 2023 – 2024

Mentor — Graduate students, undergraduate URAs, visiting students, Shalek Lab 2021 – 2025

Mentored international collaborators (Human Cell Atlas, Child Health Research Foundation Bangladesh) on single-cell analysis

Head Mentor — APISPAM (Asian Pacific Islander Sponsor Program), Harvey Mudd College 2017 – 2020

Recruited and trained 20+ mentors; organized retreat and year-long cultural events

Tutor & Grader — Harvey Mudd College (Discrete Math, CompBio, CS, ML, Databases) 2017 – 2020

Academic Excellence Program Facilitator — Harvey Mudd College Biology 2019 – 2020